Understanding Momentum Through a First-order Filter Design Perspective

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Abstract

Momentum in stochastic gradient learning has commonly been described as tracking the *first moment* of gradients. In this paper, we argue that this perspective does not fully explain momentum-based learning algorithms. Specifically, we argue that commonly-used momentum-based parameter updates, algorithms such as Polyak's Heavy-Ball and Nesterov Accelerated Gradient methods, involve a first-order *lowpass filter* acting on the stochastic gradient to dynamically regularize and reshape the learning algorithm through the location of the pole and zero of the lowpass filter. In addition, we can derive iteration-dependent learning rates that normalize the filtered gradient to satisfy a fixed trust-region constraint on the parameter update step. This viewpoint unifies popular momentum methods under a common first-order filtering abstraction and motivates a modular design strategy for the learning algorithm based on gradient filter design and trust-region-based learning-rate selection.

1 Interpreting Momentum in Stochastic Gradient Learning

Classical momentum methods like Polyak's Heavy Ball (PHB) [1, 2] and Nesterov's Accelerated Gradient (NAG) [3, 4] and the more recent Adaptive moment estimation (Adam) [5] remain canonical baselines in large-scale machine learning and deep neural network training of model parameters. Despite their practical successes, the dominant way that *momentum* is often explained as estimating and exploiting the first moment of stochastic gradients does not fully explain how accelerated learning happens in momentum-based algorithms.

In this paper, we argue that **the *momentum* component in stochastic gradient algorithms is gradient filtering, specifically using a first-order lowpass filter on the stochastic gradient**, and that this perspective provides richer insights into the behavior of the learning algorithm. Our position places the momentum component squarely within classical signal processing [6–8]. The pole and zero of the filter [9, 10] can be designed to smooth, regularize, and reshape the dynamics of the learning algorithm. This offers a principled perspective into the behavior of different momentum variants.

Historically, *momentum* was introduced as a way to convert one-step gradient methods into multistep iterative schemes [1–3, 11]. Early analyses framed these constructions through analogies with physical

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inertia [1, 12], conjugate-gradient methods [13–16], and, more recently, first-order moment estimation [5]. Parallel theoretical efforts have analyzed momentum through robust control [17, 18], Lyapunov methods [17, 19], optimal control [20], and continuous-time or stochastic differential equation (SDE) approximations [20–22]. While each framework provides valuable insights, none directly offers a unified and comprehensive interpretation that reflects the discrete, iteration-level operation of such learning algorithms.

Our central claim is that a literal mathematical interpretation of momentum already exists through classical signal-processing tools related to first-order *lowpass filters*. Rather than viewing momentum primarily through the optimization behavior it induces under a particular problem setup, we identify the momentum computation itself as a gradient-processing mechanism. Starting from a general first-order lowpass filter acting on the stochastic gradient sequence in Section 2, we show that classical momentum methods can be recovered as particular pole-zero realizations of the same filter family. The pole and zero locations determine how gradient information is attenuated, delayed, amplified, smoothed, and regularized across iterations. Under this interpretation, PHB and NAG emerge as distinct but elementary realizations within a common first-order lowpass filter family, differing primarily in the placement of a single zero relative to a stable pole.

Beyond recovering these classical schemes, in Section 3 we show that this first-order filter structure leads to complementary interpretations in terms of gradient regularization, finite-difference extrapolation, and we also derive transient bias-correction factors for the pole-zero family. To illustrate these ideas in a transparent setting, Section 4 studies a deliberately minimal one-dimensional stochastic error model. Although highly simplified, the model permits its mean-error dynamics and mean-squared error dynamics to be written explicitly as linear iteration-invariant systems. This allows closed-form joint analyses of stability, convergence, and parameter selection. Within this illustrative setting, both stability and convergence properties reduce to familiar conditions from linear systems theory. In particular, we show in this setting that the fastest mean-error and mean-square error convergence can both be achieved without gradient filtering. This underscores that acceleration is not the defining feature of momentum. Rather, momentum can be understood directly from its update equations as a lowpass regularization mechanism that reshapes the parameter-update process through filter design.

In Section 5, we further compare this first-order filter-design interpretation with several established perspectives on momentum. We argue that these perspectives primarily explain the optimization dynamics induced by momentum, whereas the present viewpoint identifies the discrete-time gradient-processing mechanism that generates those dynamics. In particular, the behavior of classical momentum schemes, including Nesterov's parameterization, follows directly from the pole-zero structure of the underlying first-order filter, rather than from implicit acceleration mechanisms invoked in continuous-time interpretations. We then study in Section 6 how alternative design choices of the zero location within the first-order lowpass filter family affect the tradeoff between variance reduction and the responsiveness of the filtered update equation to changes in the optimization landscape.

Finally, in Section 7, we conclude this paper and show that this signal-processing perspective cleanly separates optimizer design into two complementary components: gradient filtering through momentum and gradient normalization through learning-rate selection. Within this framework, gradient-dependent learning-rate rules, including those underlying stochastic gradient optimizers like Adam and Muon, admit natural trust-region interpretations [23, 24]. More broadly, this modular decomposition provides a unified framework through which existing optimizers can be compared, interpreted and new optimization algorithms can be systematically constructed.

Main Contributions: The goal of this paper is not to establish the empirical superiority of a particular optimizer, but to establish a clean and modular design language for understanding, comparing, and constructing momentum methods. Specifically, this paper advances four main contributions. **First**, we show that classical momentum methods can be unified within a common first-order pole-zero lowpass filter framework, with Heavy-Ball and Nesterov momentum arising as particular filter realizations. **Second**, we show that this first-order lowpass filter structure directly admits complementary interpretations as gradient smoothing, finite-difference-based regularization, and gradient extrapolation, and we derive the generalized transient bias-correction factor associated with this pole-zero filter family. **Third**, we show for an illustrative stochastic error model that its fastest mean-square error convergence is attained without momentum, while the fastest mean-error convergence yields the zero location of the first-order filter corresponding to Nesterov momentum. These results motivate a distinction between gradient filtering and acceleration. **Fourth**, we show that viewing momentum as gradient filtering cleanly decomposes optimizer design into two modular components: gradient filtering through filter design and trust-region-based gradient normalization through learning-rate selection.

2 Gradient Filtering

Many existing perspectives explain the behavior of optimization algorithms that use momentum under particular problem setups. As a result, the properties of the momentum mechanism itself are often intertwined with the properties of the optimization problem on which it is analyzed.

Our goal is different. Here, we instead identify the momentum computation itself. Starting directly from the stochastic gradient update equation and a general first-order filter acting on the stochastic gradient sequence, we show that classical momentum methods can be recovered as particular pole-zero realizations within a family of first-order lowpass filters constrained to have spectral norm less than one. Under this view, both Heavy-ball and Nesterov momentum emerge as distinct pole-zero designs within the same first-order lowpass filter family.

2.1 Stochastic Gradient Update

Let $\mathrm{w}(t),\ \mathrm{g}(t),\ \mathrm{v}(t),\ \mathrm{d}(t+1),\ \alpha(t) \in \mathbb{R}^{n\times 1}$, where $n \geq 1$. Let the shorthand notation (t,i) denote the i-th element of an n-dimensional vector at iteration t. Expressed coordinate-wise, the stochastic gradient update algorithm is

$$\mathrm{w}(t+1,i) = \mathrm{w}(t,i) - \alpha(t,i)\,\mathrm{g}(t,i), \tag{1}$$

where $\mathrm{w}(t,i)$ is the i-th coordinate of the parameter vector at the current iteration, $\mathrm{g}(t,i)$ is the corresponding stochastic gradient entering the update equation from a gradient-generating oracle, and $\alpha(t,i) > 0$ is the possibly iteration-dependent learning rate applied to that coordinate, obtained from a learning rate oracle. Further, we define the update step as

$$\mathrm{d}(t+1) = \mathrm{w}(t+1) - \mathrm{w}(t). \tag{2}$$

2.2 Assumptions

Throughout, we use $\mathbb{E}[\cdot]$ to denote expectation with respect to the underlying stochastic process generating the gradients. Except otherwise stated, no specific objective function, probabilistic structure, or distribution of the gradients is assumed.

Assumption 1 (Lipschitz Regularity): The function $f(t)$ is at least twice continuously differentiable, and both $f(t)$ and $\mathrm{g}(t,i)$ are Lipschitz continuous in $\mathrm{w}(t,i)$ [25].

Assumption 2 (Bounded p-th Moment): The stochastic signal $\mathrm{g}(t,i)$ satisfies $||\mathrm{g}(t,i)||_\infty < \infty$. Hence, for every finite $p \geq 1$, $\mathbb{E}[|\mathrm{g}(t,i)|^p] \leq ||\mathrm{g}(t,i)||_\infty^p < \infty$.

Assumption 3 (Bounded Step-size): Each step of the learning algorithm is constructed so that $\mathbb{E}[|\mathrm{d}(t+1,i)|^p] \leq \mu^p$, where $0 < \mu < \infty$, denotes a trust-region constant defining the maximum allowable step-size of the update step.

Next, instead of directly using the stochastic gradient $\mathrm{g}(t)$ in (1), we replace it with a lowpass filtered version. By attenuating high-frequency fluctuations in its input signal, a lowpass filter preserves low-frequency behavior and reduce sensitivity to rapid iteration-to-iteration variations.

2.3 Spectral Regularization via a First-order Lowpass Filter

Let $\mathbb{H}\colon \ell^2 \to \ell^2$ be a first-order, linear iteration-invariant filter characterized by a pole location $\beta \in \mathbb{R}$ and zero location $\gamma \in \mathbb{R}$, with transfer function

$$\mathbb{H}(z) = \eta\frac{1-\gamma z^{-1}}{1-\beta z^{-1}}, \qquad \eta = \frac{1-\beta}{1-\gamma}, \tag{3}$$

where $z = e^{j\omega}$ over frequencies $\omega \in [-\pi, \pi]$, $\mathbb{H}(1) = 1$ for unit steady-state (zero-frequency) gain, and z^{-1} denotes a one-step delay operator. The squared magnitude of the frequency response is

$$|\mathbb{H}(e^{j\omega})|^2 = \frac{(1-\beta)^2}{(1-\gamma)^2}\frac{1+\gamma^2-2\gamma\cos\omega}{1+\beta^2-2\beta\cos\omega}. \tag{4}$$

An important performance measure of robustness for $\mathbb{H}$ is its worst-case energy gain [26], namely the maximum singular value of the transfer matrix corresponding to $\mathbb{H}$. This quantity is given by the $\mathcal{H}_\infty$ norm defined as $\|\mathbb{H}\|_\infty = \max_\omega \|\mathbb{H}(e^{j\omega})\|_2 = \max_\omega |\mathbb{H}(e^{j\omega})|, \quad \forall\, \omega \in [-\pi, \pi]$ [27]. Accordingly, we restrict attention to a family of linear iteration-invariant first-order filters satisfying the design specification $\|\mathbb{H}\|_\infty \leq 1$. To show that $\|\mathbb{H}\|_\infty \leq 1$, it suffices to show conditions for which the denominator of (4) is always greater than or equal to its numerator. Rearranging terms, this implies

$$\begin{aligned}\Delta(\omega) &= (1-\gamma)^2(1+\beta^2-2\beta\cos\omega) - (1-\beta)^2(1+\gamma^2-2\gamma\cos\omega) \geq 0\\ &= 2(\beta-\gamma)(1-\beta\gamma)(1-\cos\omega) \geq 0.\end{aligned} \tag{5}$$

Note that in (5), the following holds: $1-\cos\omega \geq 0$ is non-negative for all ω; factor $\beta - \gamma \geq 0$ if $|\gamma| < |\beta|$; and factor $1 - \beta\gamma > 0$ if $|\beta| < 1$ and $|\gamma| < 1$. Therefore, a sufficient condition for both lowpass filtering and $\|\mathbb{H}\|_\infty \leq 1$ is to require the operating parameter ranges

$$0 \leq \beta < 1, \qquad |\gamma| < |\beta|. \tag{6}$$

By maintaining a spectral norm $\|\mathbb{H}\|_\infty \leq 1$, $\forall\, \omega \in [-\pi, \pi]$, this design condition (6) provides attenuation of high-frequency spectral components in the $\mathrm{g}(t,i)$ sequence, and ensures that the linear operator is non-expansive in the ℓ_2 norm. Define $\mathrm{v}(t,i) = \mathbb{H}\{\mathrm{g}(t,i)\}$. By definition of the induced norm [26, 27], this design implies $\|\mathrm{v}(t,i)\|_2 \leq \|\mathrm{g}(t,i)\|_2$, ensuring that for any deterministic realization of $\mathrm{g}(t,i)$, the total finite energy in $\mathrm{v}(t,i)$ cannot exceed the total finite energy in $\mathrm{g}(t,i)$. Since this system bound holds for all possible realizations, it directly follows that $\mathbb{E}[\mathrm{v}^2(t,i)] \leq \mathbb{E}[\mathrm{g}^2(t,i)]$.

Further, applying the convolution property of all linear iteration-invariant systems, it immediately follows that $\mathrm{v}(t,i) = \sum_{k=0}^{t} \hbar(k)\, \mathrm{g}(t-k,i)$, where $\hbar(k)$ is the impulse response obtained via the inverse z-transform of $\mathbb{H}(z)$ [6, 9], and defined as $\hbar(0) = \eta$ and $\hbar(k) = \eta(\beta-\gamma)\beta^{k-1}$ if $k > 0$. Because (6) is satisfied, it follows that $\hbar(k) \geq 0$ for all $k \geq 0$, and exponentially decaying to 0 as $k \to \infty$. Consequently, $\|\hbar\|_1 = \sum_{k=0}^{\infty} |\hbar(k)| = 1 = \mathbb{H}(1)$, and hence $\|\mathrm{v}(t,i)\|_\infty \leq \|\mathrm{g}(t,i)\|_\infty$. Therefore, by definition, the linear iteration-invariant operator is also non-expansive in its induced ℓ_∞ norm. Using Assumptions 2 and 3, if $\mathbb{E}[|\mathrm{d}(t+1,i)|^p] = \alpha^p(t,i)\,\mathbb{E}[|\mathrm{g}(t,i)|^p] \leq \alpha^p(t,i)\,\|\mathrm{g}(t,i)\|_\infty^p \leq \mu^p$ is satisfied by the unfiltered update step, then after filtering, $\mathbb{E}[|\mathrm{d}(t+1,i)|^p] = \alpha^p(t,i)\,\mathbb{E}[|\mathrm{v}(t,i)|^p]$ leads to

$$\mathbb{E}[|\mathrm{d}(t+1,i)|^p] \leq \alpha^p(t,i)\,\|\mathrm{v}(t,i)\|_\infty^p \leq \alpha^p(t,i)\,\|\mathrm{g}(t,i)\|_\infty^p \leq \mu^p. \tag{7}$$

Therefore, the designed first-order lowpass filtered update step preserves, and cannot increase the underlying maximum step-size bound satisfied by the unfiltered update step.

2.4 Canonical Filter Realization

A direct difference equation realization of the first-order filter (3) is

$$\mathrm{v}(t,i) = \beta\,\mathrm{v}(t-1,i) + \eta\,\big(\mathrm{g}(t,i) - \gamma\,\mathrm{g}(t-1,i)\big). \tag{8}$$

In general, (8) admits the controllable canonical realization [9, 10],

$$\begin{aligned}\mathrm{q}(t,i) &= \beta\,\mathrm{q}(t-1,i) + \mathrm{g}(t,i),\\ \mathrm{v}(t,i) &= \eta\big(\mathrm{q}(t,i) - \gamma\,\mathrm{q}(t-1,i)\big),\end{aligned} \tag{9}$$

where $\mathrm{q}(t,i)$ is the state generated during realization of the filter.

To make the lowpass regularization of $\mathrm{g}(t,i)$ explicit, define $\tilde{\mathrm{q}}(t,i) = (1-\beta)\,\mathrm{q}(t,i)$. Since $\eta = \frac{1-\beta}{1-\gamma}$ and $1-\eta = \frac{\beta-\gamma}{1-\gamma}$, it follows that $\eta(\beta-\gamma)\mathrm{q}(t-1,i) = (1-\eta)\tilde{\mathrm{q}}(t-1,i)$. Substituting into (9) yields

$$\begin{aligned}\tilde{\mathrm{q}}(t,i) &= \beta\,\tilde{\mathrm{q}}(t-1,i) + (1-\beta)\,\mathrm{g}(t,i),\\ \mathrm{v}(t,i) &= \eta\,\mathrm{g}(t,i) + (1-\eta)\,\tilde{\mathrm{q}}(t-1,i),\end{aligned} \quad\Leftrightarrow\quad \begin{aligned}\mathrm{e}(t,i) &= \beta\,\mathrm{e}(t-1,i) + \big(\mathrm{g}(t-1,i) - \mathrm{g}(t,i)\big),\\ \mathrm{v}(t,i) &= \mathrm{g}(t,i) + (1-\eta)\,\mathrm{e}(t,i),\end{aligned} \tag{10}$$

where the right-hand-side representation follows from applying the change of variables $\mathrm{e}(t,i) = \tilde{\mathrm{q}}(t-1,i) - \mathrm{g}(t,i)$ to the left-hand-side representation.

2.5 Recovering Momentum from the First-Order Filter Design Condition

More generally, under the design condition (6), given a stable pole location $0 \leq \beta < 1$, appropriate choices of the zero location γ allow the first-order filter to recover both Heavy-Ball and Nesterov momentum.

2.5.1 Heavy-ball and Nesterov momentum

Algebraically, analyzing (9), Heavy-ball momentum can be recovered by selecting $\gamma_{\text{HB}} = 0$,

$$\mathrm{v}(t,i) = \beta\,\mathrm{v}(t-1,i) + (1-\beta)\,\mathrm{g}(t,i), \tag{11}$$

Equivalently, the change of variables $\tilde{\mathrm{v}}(t,i) = \mathrm{v}(t,i)/(1-\beta)$, and $\tilde{\alpha}(t,i) = (1-\beta)\,\alpha(t,i)$ yields

$$\begin{aligned} \tilde{\mathrm{v}}(t,i) &= \beta\,\tilde{\mathrm{v}}(t-1,i) + \mathrm{g}(t,i) \\ \mathrm{w}(t+1,i) &= \mathrm{w}(t,i) - \tilde{\alpha}(t,i)\,\tilde{\mathrm{v}}(t,i). \end{aligned} \tag{12}$$

For non-zero γ values, from (9), $\mathrm{v}(t,i) = \eta(\beta-\gamma)\,\mathrm{q}(t-1,i) + \eta\,\mathrm{g}(t,i)$. Nesterov momentum corresponds to substituting $\gamma_{\text{NAG}} = \frac{\beta}{1+\beta}$, so that $\eta = (1-\beta)(1+\beta)$, $\eta(\beta-\gamma) = \beta^2(1-\beta)$ to obtain

$$\begin{aligned} \tilde{\mathrm{q}}(t,i) &= \beta\,\tilde{\mathrm{q}}(t-1,i) + (1-\beta)\,\mathrm{g}(t,i) \\ \mathrm{v}(t,i) &= \beta\,\tilde{\mathrm{q}}(t,i) + (1-\beta)\,\mathrm{g}(t,i). \end{aligned} \tag{13}$$

and equivalently,

$$\begin{aligned} \mathrm{q}(t,i) &= \beta\,\mathrm{q}(t-1,i) + \mathrm{g}(t,i) \\ \tilde{\mathrm{v}}(t,i) &= \beta\,\mathrm{q}(t,i) + \mathrm{g}(t,i) \\ \mathrm{w}(t+1,i) &= \mathrm{w}(t,i) - \tilde{\alpha}(t,i)\,\tilde{\mathrm{v}}(t,i). \end{aligned} \tag{14}$$

Therefore, directly from the update equations, *momentum* in stochastic gradient algorithms emerges as a first-order lowpass filtering operation on the stochastic gradient.

3 Lowpass Regularization in the Iteration-domain

From (10), recall $\mathrm{e}(t,i) = \beta\,\mathrm{e}(t-1,i) + \big(\mathrm{g}(t-1,i) - \mathrm{g}(t,i)\big)$. Using the unit-delay operator z^{-1}, and defining the first-order finite-difference operator $\Delta = 1 - z^{-1}$, we have that $\big(\mathrm{g}(t,i) - \mathrm{g}(t-1,i)\big) = (1-z^{-1})\mathrm{g}(t,i) = \Delta\mathrm{g}(t,i)$. The $\mathrm{e}(t,i)$ state recursion with zero initialization $\mathrm{e}(0) = 0$, and repeated substitutions can then be expressed as

$$\mathrm{e}(t,i) = -\sum_{k=0}^{t-1} \beta^k z^{-k}(1-z^{-1})\mathrm{g}(t,i) = -\sum_{k=0}^{t-1} \beta^k z^{-k}\Delta\mathrm{g}(t,i), \tag{15}$$

and hence $\mathrm{v}(t,i) = \mathrm{g}(t,i) + (1-\eta)\,\mathrm{e}(t,i)$ is

$$\mathrm{v}(t,i) = \mathrm{g}(t,i) - (1-\eta)\sum_{k=0}^{t-1} \beta^k z^{-k}\Delta\mathrm{g}(t,i). \tag{16}$$

Equation (15) is an exponentially weighted sum of the delayed first-order finite differences of $\mathrm{g}(t,i)$. Each term $z^{-k}(1-z^{-1})\mathrm{g}(t,i)$, is generated by delayed applications of the first-order finite-difference operator $(1-z^{-1})$ on $\mathrm{g}(t,i)$, where $z^{-k}\mathrm{g}(t,i) = \mathrm{g}(t-k,i)$, and $z^{-k}\Delta\mathrm{g}(t,i) = (1-z^{-1})\mathrm{g}(t-k,i)$. Hence, (16) can be interpreted as the gradient component $\mathrm{g}(t,i)$ being regularized through a correction term formed from an exponentially weighted sum of its delayed first-order finite differences.

Consequently, beyond typical smoothing, $\mathbb{H}$ can be viewed as performing a *regularization* of the gradient sequence by selectively penalizing large iteration-to-iteration changes in each co-ordinate of the gradient $\mathrm{g}(t)$.

Further, since $z^{-1} = 1 - \Delta$, it follows that $z^{-k} = (1-\Delta)^k$. Applying the Newton (generalized binomial) expansion,

$$z^{-k} = (1-\Delta)^k = \sum_{r=0}^{\infty} (-1)^r \binom{k}{r} \Delta^r, \tag{17}$$

leads to

$$\mathrm{e}(t,i) = \sum_{k=0}^{t-1} \beta^k \sum_{r=0}^{\infty} (-1)^{r+1} \binom{k}{r} \Delta^{r+1}\mathrm{g}(t,i). \tag{18}$$

Next, interchange the summations (justified by the absolute convergence of the series for $|\beta| < 1$),

$$\mathrm{e}(t, i) = \sum_{r=0}^{\infty}(-1)^{r+1}\left(\sum_{k=0}^{t-1}\binom{k}{r}\beta^k\right)\Delta^{r+1}\mathrm{g}(t, i). \tag{19}$$

Note that by definition, the binomial coefficient $\binom{k}{r} = 0$ whenever $r > k$, and $k \leq t-1$, therefore every term where $r \geq t$ and $k < r$ evaluates to zero. Therefore, the upper index of the external infinite sum intrinsically truncates at $t-1$, that is $0 \leq r \leq t-1$. Similarly, the lower index of the inner sum can be tightened to start at $k = r$, so that $r \leq k \leq t-1$, the equation then becomes

$$\mathrm{e}(t, i) = \sum_{r=0}^{t-1}(-1)^{r+1}\left(\sum_{k=r}^{t-1}\binom{k}{r}\beta^k\right)\Delta^{r+1}\mathrm{g}(t, i), \tag{20}$$

Finally, re-indexing the external sum with $j = r+1$, and defining the iterative penalty weight

$$\pi_j(\beta, t) = (-1)^j \sum_{k=j-1}^{t-1}\binom{k}{j-1}\beta^k, \tag{21}$$

we have the compact form

$$\mathrm{e}(t, i) = \sum_{j=1}^{t}\pi_j(\beta, t)\,\Delta^j\mathrm{g}(t, i), \tag{22}$$

and therefore $\mathrm{v}(t, i) = \mathrm{g}(t, i) + (1-\eta)\,\mathrm{e}(t, i)$ can also be expressed in the finite-difference form

$$\mathrm{v}(t, i) = \mathrm{g}(t, i) + (1-\eta)\sum_{j=1}^{t}\pi_j(\beta, t)\,\Delta^j\mathrm{g}(t, i). \tag{23}$$

Under this representation (22), the correction variable $\mathrm{e}(t, i)$ admits an alternating finite-difference expansion involving finite differences up to order t. Odd-order finite differences related to sharp variations are penalized negatively, whereas even-order finite differences related to smooth variations are penalized positively.

Consequently, the filter output may be interpreted not only as an estimate of $g(t, i)$ with suppressed high-frequency fluctuations, but also as an extrapolation of $\mathrm{g}(t, i)$ via a finite-difference-based correction term composed of t penalized higher-order finite-differences of $\mathrm{g}(t, i)$.

3.1 Transient Bias Correction

The finite-difference representation in (23) also provides a direct characterization of the filter's transient response. In particular, it allows the discrepancy between the finite-time output $\mathrm{v}(t, i)$ and its asymptotic value $\mathrm{v}(\infty, i)$ to be computed exactly. To quantify this effect, consider a constant input

$$\mathrm{g}(k, i) = \begin{cases} 0, & k \leq 0, \\ c, & k > 0. \end{cases}$$

The finite differences from (23), can be evaluated explicitly. Since $\Delta^j = (1 - z^{-1})^j = \sum_{m=0}^{j}(-1)^m\binom{j}{m}z^{-m}$, we obtain $\Delta^j\mathrm{g}(t, i) = \sum_{m=0}^{j}(-1)^m\binom{j}{m}\mathrm{g}(t-m)$.

For $j < t$, every delayed sample satisfies $\mathrm{g}(t-m, i) = c$, hence using the binomial identity

$$\Delta^j\mathrm{g}(t, i) = c\sum_{m=0}^{j}(-1)^m\binom{j}{m} = c(1-1)^j = 0.$$

For $j = t$, $\Delta^t\mathrm{g}(t, i) = c\sum_{m=0}^{t-1}(-1)^m\binom{t}{m}$, using $\sum_{m=0}^{t}(-1)^m\binom{t}{m} = (1-1)^t = 0$, the truncated sum is

$$\sum_{m=0}^{t-1}(-1)^m\binom{t}{m} = -(-1)^t,$$

giving $\Delta^t \mathrm{g}(t,i) = c(-1)^{t-1}$. Substituting into the finite-difference expansion leaves only the boundary term, $\mathrm{v}(t,i) = c + (1-\eta)(-1)^t \pi_j(\beta,t) c(-1)^{t-1}$. Since $\pi_j(\beta,t) = \binom{t-1}{t-1}\beta^{t-1} = \beta^{t-1}$, it follows that

$$\mathrm{v}(t,i) = c\big[1-(1-\eta)\beta^{t-1}\big].$$

The steady-state value is obtained by taking $t \to \infty$, yielding $\mathrm{v}(\infty,i) = c$, and therefore the transient bias is $b(t) = \mathrm{v}(\infty,i) - \mathrm{v}(t,i) = c(1-\eta)\beta^{t-1}$. Consequently, an unbiased estimate of the steady-state response is obtained by the exact correction

$$\hat{\mathrm{v}}(t,i) = \frac{\mathrm{v}(t,i)}{1-(1-\eta)\beta^{t-1}}. \tag{24}$$

For Heavy-Ball momentum ($\gamma = 0$), $1-\eta = \beta$, recovering the familiar correction $\hat{\mathrm{v}}(t,i) = \frac{\mathrm{v}(t,i)}{1-\beta^t}$. Similarly, for Nesterov momentum ($\gamma = \beta/(1+\beta)$), $1-\eta = \beta^2$, leading to $\hat{\mathrm{v}}(t,i) = \frac{\mathrm{v}(t,i)}{1-\beta^{t+1}}$. Thus, the commonly used bias correction appears here as a special case of a more general result applicable to the entire pole-zero family considered in this work.

4 An Illustrative Stochastic Error Model

In this section, we study what momentum does in one of the simplest stochastic optimization setups. In analyzing momentum, the use of simplified scalar models as a tractable environment free of multidimensional complications is well established in the literature [15, 17, 28–30]. We also discover, in the simplest possible setting, how the zero location of the filter corresponding to Nesterov momentum can be derived.

Modeling setup: Consider the simple model of minimizing, at each iteration $t > 0$, a twice-differentiable, weighted stochastic error function,

$$f(t) = \tfrac{1}{2}\,\psi^2(t)\,\cdot\varepsilon^2(t) \tag{25}$$

with respect to the model parameter $w(t) \in \mathbb{R}$, whose error relative to an optimum $w^* \in \mathbb{R}$ is $\varepsilon(t) = w(t) - w^*$, and weighting variable $\psi(t)$ statistically independent of the error $\varepsilon(t)$, modeled as a realization of a wide-sense stationary, stochastic data process with zero mean $\mathbb{E}[\psi(t)] = 0$, a finite second-moment $\mathbb{E}[\psi^2(t)] = \lambda$, fourth-moment $\mathbb{E}[\psi^4(t)] = \kappa\lambda^2$ and uncorrelated values across time $\mathbb{E}[\psi(t)\,\psi(k)] = 0$, for $t \neq k$, for which its variance $0 < \lambda < \infty$ and kurtosis $1 \leq \kappa < \infty$ exists.

Differentiating $f(t)$ with respect to $\mathrm{w}(t)$, we get the associated one-dimensional scalar gradient $g(t) = \psi^2(t)\,\varepsilon(t)$, and Hessian $h(t) = \psi^2(t)$. Taking expectations,

$$\mathbb{E}[f(t)] = \lambda\,\mathbb{E}[\varepsilon^2(t)], \quad \mathbb{E}[g(t)] = \lambda\,\mathbb{E}[\varepsilon(t)], \quad \mathbb{E}[h(t)] = \lambda, \quad \mathbb{E}[g^2(t)] = \kappa\lambda^2\mathbb{E}[\varepsilon^2(t)]. \tag{26}$$

In this setup, $g(t) = \psi^2(t)\,\varepsilon(t)$ involves multiplicative stochasticity, and is structurally analogous to a large class of stochastic approximation and adaptive filtering analyses in which a random data process multiplies the parameter error [7, 31–33], and in which stochastic error surfaces are modeled through time-varying data processes. Consequently, this setup should be viewed as an analytically tractable model for studying learning dynamics under multiplicative gradient noise. In the following, we will see that ensuring convergence within this problem setup is equivalent to certifying the stability conditions of a linear iteration-invariant system [34]. We will also see that neither its fastest mean square error nor its fastest mean-error convergence, necessarily requires momentum [7].

Further, denote $v(t) = \mathbb{H}_{\beta,\gamma}\{g(t)\}$ as the smoothed gradient in this setup, and assume a constant learning-rate $\alpha(t) = \alpha$, so that the update equation can be written as $\varepsilon(t+1) = \varepsilon(t) - \alpha\,v(t)$ or equivalently as $\varepsilon(t+1) = \varepsilon(t) - \tilde{\alpha}\,\tilde{v}(t)$ via change of variables $\tilde{\alpha} = (1-\beta)\,\alpha$, and $\tilde{v}(t) = v(t)/(1-\beta)$.

4.1 Mean-square error convergence

From (26), minimizing the mean-square error $\mathbb{E}[\varepsilon^2(t)]$ is equivalent to minimizing both the expected loss $\mathbb{E}[f(t)]$ and the gradient's second-moment $\mathbb{E}[g^2(t)]$. Our objective is therefore to determine the constant parameterization (α, β, γ) that yields the fastest mean-square error convergence.

Applying the canonical filter realization

$$\begin{aligned} q(t) &= \beta\, q(t-1) + g(t) \\ v(t) &= \eta\big(q(t) - \gamma\, q(t-1)\big) \\ \varepsilon(t+1) &= \varepsilon(t) - \alpha\, v(t). \end{aligned} \tag{27}$$

For convenience, let $a = \alpha\,\eta$, and $\delta = \beta - \gamma$. Substituting $g(t) = \psi^2(t)\varepsilon(t)$ gives

$$\begin{aligned} q(t) &= \beta\, q(t-1) + \psi^2(t)\varepsilon(t) \\ \varepsilon(t+1) &= \big(1 - a\psi^2(t)\big)\varepsilon(t) - a\delta q(t-1). \end{aligned} \tag{28}$$

Working out the algebra of the recursion and using (26) leads to

$$\mathbb{E}[\varepsilon^2(t+1)] = (1 - 2a\lambda + a^2\lambda^2\kappa)\mathbb{E}[\varepsilon^2(t)] - 2a\delta(1 - a\lambda)\mathbb{E}[\varepsilon(t)q(t-1)] + a^2\delta^2\mathbb{E}[q^2(t-1)] \tag{29}$$

$$\mathbb{E}[\varepsilon(t+1)\, q(t)] = (\beta(1 - a\lambda) - a\lambda\delta)\mathbb{E}[\varepsilon(t)q(t-1)] + \lambda(1 - a\kappa\lambda)\mathbb{E}[\varepsilon^2(t)] - a\beta\delta\, \mathbb{E}[q^2(t-1)] \tag{30}$$

$$\mathbb{E}[q^2(t)] = \beta^2\, \mathbb{E}[q^2(t-1)] + 2\beta\lambda\, \mathbb{E}[\varepsilon(t)q(t-1)] + \kappa\lambda^2\mathbb{E}[\varepsilon^2(t)]. \tag{31}$$

Define $\mathrm{y}(t+1) = [\mathbb{E}[\varepsilon^2(t+1)], \mathbb{E}[\varepsilon(t+1)q(t)], \mathbb{E}[q^2(t)]]^\top \in \mathbb{R}^{3\times 1}$ as state variable vector. Therefore, in matrix form, the mean-square error dynamics, with its covariance matrix $\mathbf{B}$ is

$$\mathrm{y}(t+1) = \underbrace{\begin{bmatrix} 1 - 2a\lambda + a^2\lambda^2\kappa & -2a\delta(1 - a\lambda) & a^2\delta^2 \\ \lambda(1 - a\kappa\lambda) & \beta(1 - a\lambda) - a\lambda\delta & a\beta\delta \\ \kappa\lambda^2 & 2\beta\lambda & \beta^2 \end{bmatrix}}_{\mathbf{B}} \mathrm{y}(t) \tag{32}$$

The cross-moment dynamics (30) entering the mean-square error (29) is a filtering system driven by two non-negative energy quantities. To achieve the smallest possible cross-term, for any initialization of the cross-moment term, we first need to drive its input contribution to zero. Inspecting (30), this requires $a\delta = 0$ and $\lambda(1 - a\kappa\lambda) = 0$. Since $\delta = \beta - \gamma$, $a = \alpha\eta > 0$, $\lambda > 0$, and $\kappa > 0$, this implies that the unique solution that removes the cross-term coupling for constant (α, β, γ) is

$$\boxed{\gamma = \beta, \quad \alpha = \tfrac{1}{\eta\,\kappa\lambda}}. \tag{33}$$

Moreover, differentiating (29) with respect to a and setting to 0, $d\,\mathbb{E}[\varepsilon^2(t+1)]/da = -\lambda(1 - a\kappa\lambda)\mathbb{E}[\varepsilon^2(t)] - \delta(1 - 2a\lambda)\mathbb{E}[\varepsilon(t)q(t-1)] + a\delta\,\mathbb{E}[q^2(t-1)] = 0$. By inspection, the stationary condition is achieved for constant coefficients if and only if $\delta = 0$ and $\lambda(1 - a\kappa\lambda) = 0$, leading to (33).

Under this choice (33), $\eta = \frac{1-\beta}{1-\gamma} = 1$, and $\mathbb{H}_{\beta,\gamma}(z) = \frac{1-\beta z^{-1}}{1-\beta z^{-1}} = 1$. Consequently, the filter collapses to the identity operator, and $v(t) = g(t)$. Substituting (33) into (32) yields a lower triangular covariance matrix

$$\mathrm{y}(t+1) = \underbrace{\begin{bmatrix} 1 - \frac{1}{\kappa} & 0 & 0 \\ 0 & \beta(1 - \frac{1}{\kappa}) & 0 \\ \kappa\lambda^2 & 2\beta\lambda & \beta^2 \end{bmatrix}}_{\mathbf{B}} \mathrm{y}(t). \tag{34}$$

Therefore, its eigenvalues are its diagonal entries,

$$z_1 = 1 - \tfrac{1}{\kappa}, \quad z_2 = \beta(1 - \tfrac{1}{\kappa}), \quad z_3 = \beta^2. \tag{35}$$

At this point, the only remaining free parameter is β. Since $0 \leq \beta < 1$, both residual eigenvalues z_2 and z_3 are simultaneously minimized to $z_2 = z_3 = 0$ by the choice $\beta = 0$, transforming the lower-triangular covariance matrix to a rank-1 matrix,

$$\mathrm{y}(t+1) = \underbrace{\begin{bmatrix} 1 - \frac{1}{\kappa} & 0 & 0 \\ 0 & 0 & 0 \\ \kappa\lambda^2 & 0 & 0 \end{bmatrix}}_{\mathbf{B}} \mathrm{y}(t). \tag{36}$$

Consequently, no alternative constant parameterization (α, β, γ) can reduce the covariance matrix below this rank-one form. Therefore, the optimal constant parameterization is

$$\boxed{\alpha = \tfrac{1}{\kappa\lambda}, \quad \gamma = \beta = 0}. \tag{37}$$

Since this optimal mean-square error solution collapses the pole-zero pair, eliminating the filter entirely and reducing the algorithm to the plain stochastic gradient method $\gamma = \beta = 0$. Therefore, for this setup, the fastest mean-square error convergence does not necessarily require momentum.

4.2 Mean error convergence

Recall that classical momentum variants correspond to distinct pole–zero placements of the lowpass filter, $\mathbb{H}_{\beta,\gamma}$. Here, we will see that the filter's zero location corresponding to Nesterov momentum, yields the fastest exponential decay of the mean error.

From (26), minimizing the mean-error variable is equivalent to minimizing the first-moment of the gradient. Our objective is therefore to determine the constant parameterization (α, β, γ) that yields the fastest mean error convergence.

Similarly, define $\mathrm{x}(t+1) = [\mathbb{E}[\varepsilon(t+1)], \mathbb{E}[q(t)]]^\intercal \in \mathbb{R}^{2\times 1}$ as the state variable vector associated with the mean error,

$$\begin{aligned} \mathbb{E}[\varepsilon(t+1)] &= (1 - a\lambda)\, \mathbb{E}[\varepsilon(t)] - a\delta\, \mathbb{E}[q(t-1)] \\ \mathbb{E}[q(t)] &= \lambda\, \mathbb{E}[\varepsilon(t)] + \beta\, \mathbb{E}[q(t-1)]. \end{aligned} \tag{38}$$

Therefore, in matrix form, the mean-error dynamics is

$$\mathrm{x}(t+1) = \underbrace{\begin{bmatrix} (1 - a\lambda) & -a\delta \\ \lambda & \beta \end{bmatrix}}_{\mathbf{A}} \mathrm{x}(t) \tag{39}$$

The two-state linear, iteration invariant system (39) can be written compactly as $\mathrm{x}(t) = \mathbf{A}^t\, \mathrm{x}(0)$ [35]. Its characteristic polynomial $\det(z\mathbf{I} - \mathbf{A}) = 0$ gives $z^2 - (1 + a_p)\, z + (a_p - k_p) = 0$, where $a_p = \beta - \alpha\, \eta\, \lambda, , k_p = -\alpha\, \eta\, \lambda\, (1 - \gamma)$, and $a_p - k_p = \beta - \alpha\, \eta\, \lambda\, \gamma$. From the explicit characteristic polynomial, achieving the fastest decay possible requires the common magnitude $|z_1| = |z_2| = 0$, which can only occur when both $1 + a_p = 0$ and $a_p - k_p = 0$.

Setting $a_p - k_p = \beta - \alpha\eta\lambda\,\gamma = 0$ yields $\gamma = \frac{\beta}{\alpha\eta\lambda}$, and $1 + a_p = 1 + \beta - \alpha\eta\lambda = 0$ gives $\alpha\eta\lambda = 1 + \beta$. Therefore, the mean-error optimal choice $\alpha = \frac{1+\beta}{\eta}\frac{1}{\lambda}$ and $\gamma = \frac{\beta}{1+\beta}$ yields

$$0 \le \beta < 1, \quad \gamma = \tfrac{\beta}{1+\beta}, \quad \tilde{\alpha} = \tfrac{1}{\lambda}. \tag{40}$$

where $\tilde{\alpha} = (1 - \beta)\alpha$. This shows where the zero location of the filter, $\gamma = \frac{\beta}{1+\beta}$, corresponding to Nesterov momentum is derived from. In particular, the special case $\gamma = \beta = 0$ is subsumed by $\gamma = \frac{\beta}{1+\beta}$ which reduces to $\gamma = 0$ for $\beta = 0$.

Remarks: Here, observe that, in the absence of momentum, the mean-square error dynamics always collapses to a rank-one matrix. The intended point here is that, although neither the fastest mean-square convergence nor the fastest mean convergence requires momentum, the full-rank behavior of both the mean-square error matrix $\mathbf{B}$ and the mean-error matrix $\mathbf{A}$ requires momentum. Furthermore, rapid mean convergence achieved through the effective learning rate $a = \alpha\eta = \frac{1}{\lambda}$ does not necessarily imply improved mean-square convergence or stability properties.

Now consider (6), $0 < \beta < 1$ and $|\gamma| < |\beta|$. Noting that $|\delta| = |\beta - \gamma| \le 2|\beta|$, and using the mean-square-error-optimal effective learning rate $a = \frac{1}{\kappa\lambda}$, subtract the rank-one matrix $\mathbf{B}$ in (36) from the general matrix $\mathbf{B}(a, \beta, \delta)$ in (32) to yield a perturbation matrix $\mathbf{E}$ whose first column is identically zero, while its second and third columns consist of $O(\beta)$ and $O(\beta^2)$ perturbations, respectively. Consequently, as $\beta \to 0$, we obtain $\|\mathbf{E}\| \to 0$. It follows that the momentum hyperparameters (β, γ) can be chosen so that their mean-square error convergence approaches the fastest achievable convergence obtained without momentum, while preserving a full-rank matrix structure. For example, this can be achieved by choosing $0 < \beta \ll \frac{1}{\kappa}$ with $|\gamma| < |\beta|$. Thus, momentum still acts as a mechanism for regularizing the update-step process.

5 Alternative Views of Momentum

In the literature, momentum has been interpreted through several theoretical frameworks over the past six decades. Rather than viewing these perspectives as mutually exclusive, we view them as emphasizing different aspects of the same underlying algorithm. Our position is not that these viewpoints are incorrect or incomplete. Instead, we argue that the filter-design viewpoint provides a literal iterative description of the momentum computation itself and therefore complements existing perspectives.

First-Moment Estimation View: Perhaps the most common modern interpretation is that momentum estimates the first moment of the stochastic gradient [5]. Under this viewpoint, the momentum variable is interpreted as an exponentially weighted estimator of the first moment of the stochastic gradient. This interpretation, which also treats momentum as an exponentially weighted moving average of past gradients, partly helps to explain the smoothing effect of momentum and its robustness to gradient noise. However, it provides limited insight into why different momentum variants, such as Heavy-Ball and Nesterov momentum, exhibit different dynamical behavior despite both involving an exponential smoothing operation. From the filter-design perspective, these differences arise naturally through the placement of poles and zeros of a first-order lowpass filter.

Moreover, relating a time-averaged quantity to an underlying statistical expectation requires assumptions on the gradient-generating process. In particular, interpreting the momentum variable as an estimate of an underlying first moment relies on ergodic assumptions, under which time averages can be related to ensemble expectations [36]. In contrast, the lowpass behavior induced by the pole-zero filter structure follows directly from the update equations and provides a direct description of how a stable linear operator processes gradient information over iterations, regardless of the underlying stochastic process.

Physical Inertia Analogies: Historically, momentum has often been explained using analogies to physical inertia and damped mechanics, particularly following Polyak's Heavy-Ball method [1, 2]. These interpretations provide useful intuition regarding acceleration, and rely on analogies between the optimization problem and its mechanics under particular modeling assumptions. However, they do not directly characterize the iterative computation being applied to the gradient sequence. In contrast, our position replaces the physical analogy with a linear system that yields an exact iterative realization and from it, identifies momentum as a first-order lowpass filter acting on the gradient sequence.

Continuous-Time, ODE, and SDE Perspectives: A substantial body of work studies momentum through continuous-time ordinary differential equations (ODE), stochastic differential equations (SDE), and related dynamical-system formulations [20–22]. These approaches provide valuable asymptotic analyses and often illustrate accelerated convergence-rate insights. However, the resulting interpretations are obtained after transforming the native discrete-time algorithm into a continuous-time representation. Our position instead focuses on the native iteration-level algorithm itself. Because the analysis is formulated in discrete time, the pole-zero structure of the momentum component appears explicitly, without requiring a limiting continuous-time approximation. This makes the filtering mechanism responsible for gradient smoothing directly observable within the algorithm's update equations.

Control-Theoretic Perspectives: Another influential line of work analyzes momentum using robust control, Lyapunov functions, Integral Quadratic Constraints (IQCs), Linear Matrix Inequalities (LMIs), and optimal-control formulations [17–20]. These frameworks utilize powerful tools for proving stability, robustness, and convergence guarantees. Their primary strength lies in analyzing the behavior of the complete optimization system under a specified problem setup. Our position is complementary and operates at a different level. By analyzing the optimizer as a feedback system, control-theoretic viewpoints primarily address questions of stability, robustness, and performance guarantees under a specified problem setup. In contrast, our viewpoint identifies momentum as the smallest-order, stable signal-processing block that transforms gradient information before it enters the parameter update. Consequently, the two perspectives are complementary: one emphasizes system-level guarantees under a specified optimization setup, while the other emphasizes the structure of the gradient-processing mechanism itself.

Classic Quadratic-Minimization Perspectives: A classical perspective studies momentum in the context of strongly convex quadratic minimization. In this setting, Heavy-Ball and Nesterov methods induce second-order linear recurrences whose convergence properties can be characterized through eigenvalues, convergence factors, companion matrices, stability regions, and related linear-systems analyses [1, 2, 11]. This literature explains how momentum interacts with the structure of a particular optimization problem. The distinction is therefore one of emphasis rather than contradiction. Classical quadratic analyses typically begin from the optimization dynamics and study their resulting behavior. The filter-design viewpoint instead begins from a first-order filter and shows how Heavy-Ball and Nesterov momentum emerge as particular pole-zero realizations of that filter. Thus, the two perspectives are complementary: one explains the resulting optimization dynamics, while the other explains the gradient-processing mechanism that contributes to generating those dynamics.

Multi-Step and Conjugate-Gradient Perspectives: Momentum has also been motivated through multi-step iterative methods and linked to conjugate-gradient ideas [13–16]. Under these viewpoints, information from previous iterations is reused to construct improved update directions relative to one-step gradient methods [2, 13]. These perspectives explain how past information contributes to the update direction and influences convergence behavior in a given optimization setting. Our present viewpoint is complementary. It emphasizes the mechanism that produces this behavior by focusing on a different question: how is gradient information transformed before it is applied in the parameter update?

Frequency-Domain Perspectives: Recent work [37, 38] studies momentum from a frequency-domain perspective, analyzing how momentum shapes the spectral content of gradient signals and how these frequency-selective effects influence optimization performance under specific problem setups. One distinction is that our position is not primarily a frequency-response analysis of an existing optimizer. Rather than introducing alternative optimization heuristics, our interpretation provides a constructive view of momentum in which algorithmic variants arise naturally from alternative pole-zero realizations of a first-order lowpass filter satisfying the design condition that its spectral norm is less than one.

Summary: Existing interpretations each provide valuable insights into aspects of optimization behavior due to momentum. In this sense, our position is not that these existing perspectives are redundant, but that the iterative momentum computation itself can be directly constructed from a stable first-order lowpass filter characterized by a pole and zero. Consequently, momentum design becomes filter design, while learning-rate selection may be viewed separately as a tool for gradient normalization through trust-region principles [24]. This separation provides a natural framework for understanding, comparing, and designing momentum. More broadly, it places momentum design within the language of classical signal processing and filter theory while remaining directly connected to the algorithm's update equations.

6 Alternative Lowpass Filter Designs

A central implication of this first-order filter-design viewpoint is that momentum methods are no longer isolated algorithmic constructions, but operating points within a common pole-zero design space. In the conventional optimization literature, Heavy-Ball and Nesterov momentum are typically introduced as distinct algorithms motivated through different derivations. However, under the current first-order filter representation developed in Section 2, both arise simply as particular choices of the zero location γ relative to a stable pole location β of a first-order filter satisfying a spectral norm design condition (6). This observation has an important consequence. If Heavy-Ball and Nesterov momentum correspond to two specific points within a larger admissible design space, then there is no *a priori* reason to restrict attention exclusively to these two choices. Instead, the momentum component becomes a filter-design problem in which the pole-zero configuration may be selected according to a desired performance criterion.

From this perspective, the contribution of this filter-design language is not merely to reinterpret existing momentum methods. Rather, it provides a systematic mechanism for constructing, analyzing, and comparing alternative momentum designs. Different pole-zero placements induce the need to balance different design tradeoffs [39] between smoothing, variance reduction, responsiveness to changes in the gradient sequence, and transient behavior. To illustrate this design perspective, we now consider two example filter-design criteria for selecting γ given a stable pole location β. The first seeks the maximum achievable variance reduction. The second seeks the maximum variance reduction per unit responsiveness. These criteria lead to alternative zero placements that differ from the Heavy-Ball and Nesterov choices and demonstrate how new momentum variants can be derived directly from classical filter-design principles.

6.1 Variance reduction factor

A useful measure of the output variance is the system $\mathcal{H}_2$ norm, its average energy over all frequencies,

$$||\mathbb{H}||_2^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} ||\mathbb{H}(e^{j\omega})||_F^2 \, d\omega = \frac{1}{2\pi}\int_{-\pi}^{\pi} |\mathbb{H}(e^{j\omega})|^2 \, d\omega = \frac{1-\beta}{1+\beta}\frac{1+\gamma^2-2\beta\gamma}{(1-\gamma)^2} \tag{41}$$

In particular, for a white noise input signal, this quantifies the residual noise variance after filtering [27, 39]. Therefore, minimizing (41) can be interpreted as maximizing the filter's variance reduction strength. Given $0 \le \beta < 1$, (41) can be used to place γ. Differentiating with respect to γ, $\frac{d}{d\gamma}||\mathbb{H}||_2^2$ is strictly positive, and increasing over the admissible range $-\beta < \gamma < \beta$. Consequently, the minimum variance placement for the zero location γ is

$$\boxed{\gamma_{\text{MVR}} = -\beta}. \tag{42}$$

Substituting $\gamma = -\beta$ into $\eta = \frac{1-\beta}{1-\gamma}$, yields $\eta_{\text{MVR}} = \frac{1-\beta}{1+\beta}$. As $\beta \to 1$, $\eta_{\text{MVR}} = o(1-\beta)$. Note that if $\eta = 1-\beta$, then $\gamma = 0$. Either way, maximum variance reduction is obtained at the cost of an increasingly small η. The normalization gain η governs how the filter, realized in (10) responds to changes in its input signal. In this case, the input signal $\mathrm{g}(t,i)$ carries information about the local optimization landscape. As $\eta \to 1$, minimal variance reduction, but more responsiveness to changes in the optimization landscape. However as $\eta \to 0$, maximal variance reduction, more frequency components in the input signal are attenuated more strongly, smoothing effect is increased, and can lead to less responsiveness to meaningful changes in $\mathrm{g}(t,i)$.

Reducing $||\mathbb{H}||_2^2$ leads to a tradeoff between two competing effects. Maximum variance reduction causes a simultaneous maximum reduction in the filter's normalization gain, leading to a much smaller effective maximum step-size and less responsive dynamics. An efficient filter design criterion should balance this tradeoff by simultaneously reducing variance while preserving the ability to track changes in its input. An example is a filter design with maximal variance reduction per unit gain.

6.2 Variance reduction per unit gain

A more efficient design tradeoff is to balance the variance measure relative to its responsiveness measure, defined as

$$J(\gamma) = \frac{||\mathbb{H}||_2^2}{\eta} = \frac{1+\gamma^2-2\beta\gamma}{(1+\beta)(1-\gamma)}. \tag{43}$$

Minimizing (43) is equivalent to maximizing the filter's variance reduction per unit gain. Differentiating with respect to γ yields $(1-\gamma)^2 = 2(1-\beta)$, whose feasible root is $\gamma = 1-\sqrt{2(1-\beta)}$. Subject to $-\beta \le \gamma \le \beta$, the optimal solution becomes

$$\boxed{\gamma_{\text{VRG}} = \max\left\{-\beta,\ 1-\sqrt{2(1-\beta)}\right\}.} \tag{44}$$

The transition occurs when $-\beta = 1-\sqrt{2(1-\beta)}$, which yields a threshold $\beta = \sqrt{5}-2 \approx 0.236$ related to the golden ratio. Therefore, for small $0 \le \beta < \sqrt{5}-2$, the variance-per-unit-gain optimum $\gamma_{\text{VRG}} = -\beta$ coincides with the maximum variance reduction choice. Otherwise, as $\beta \to 1$, for $\sqrt{5}-2 \le \beta < 1$, the variance-per-unit-gain optimum increasingly favors preservation of responsiveness, $\gamma_{\text{VRG}} = 1-\sqrt{2(1-\beta)}$.

Given $0 \le \beta < 1$, four special operating points of the filter are therefore

$$\gamma_{\text{MVR}} = -\beta, \quad \gamma_{\text{HB}} = 0, \quad \gamma_{\text{NAG}} = \frac{\beta}{1+\beta}, \quad \gamma_{\text{VRG}} = \max\{-\beta, 1-\sqrt{2(1-\beta)}\}. \tag{45}$$

More specifically, within the same first-order filter family with a spectral norm less than one design condition (6), both Heavy-ball and Nesterov momentum can be viewed as specific operating choices of γ. In particular, Figure 2 shows that Nesterov's momentum achieves more variance reduction per unit gain than Heavy-ball, for $\beta \gtrapprox 0.71$. Consequently, this spectral lowpass regularization interpretation of classic momentum methods through the pole-zero locations of a first-order lowpass filter, also correspond to tradeoffs between maximal variance reduction and responsiveness of the filtering dynamics to the local optimization landscape.

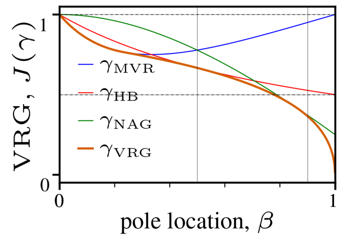

Figure 1: Variance reduction per unit gain. For $\beta > \sqrt{5} - 2$, γ_{MVR} clearly becomes inefficient. Observe that γ_{NAG} closely matches γ_{VRG} as $\beta \gg 0.5$, especially $\beta \approx 0.8 - 0.9$, whereas γ_{HB} closely matches γ_{VRG} for $\beta \approx 0.4 - 0.6$. At the cross-over point $\beta \approx 0.71$, γ_{NAG} overtakes γ_{HB} as an efficient design.

7 Conclusion: Implications of Viewing Momentum as Gradient Filtering

In this section, we articulate the conceptual implications of interpreting momentum as a first-order filtering operation, and outline how it reshapes optimizer design and tuning.

7.1 The intrinsic role of momentum is lowpass regularization.

Momentum can be fundamentally viewed as a first-order gradient-filtering mechanism. The lowpass filtering action follows directly from the update equations and is therefore present independently of the optimization problem. By contrast, acceleration is not an intrinsic property of the momentum computation itself, but a performance characteristic that may emerge through the interaction of the resulting filtered-gradient dynamics with a particular optimization landscape.

A central argument of the paper is therefore that acceleration need not be the sole justification for talking about the effect of momentum. This distinction is consistent with prior analyses in the stochastic optimization literature [7, 30] analyzing a nontrivial covariance spectrum with multiple eigenmodes and showing that gradient smoothing (or momentum) can significantly improve steady-state performance while having minimal impact on improving convergence rate. To underscore this, our analysis within the illustrative stochastic error setup prove that using momentum does not necessarily yield faster or accelerated convergence than the plain unfiltered stochastic gradient method. Thus, when the typical condition-number driven acceleration is absent, momentum still acts as a mechanism for reshaping and regularizing the parameter-update dynamics. Even when classical condition-number-driven acceleration is present, lowpass filtering is still present by construction in every momentum update. Any resulting acceleration depends on the interaction between this first-order filter and the particular optimization problem being solved.

This suggests that the intrinsic role of momentum is its lowpass regularization of the parameter-update process, while acceleration is a possible consequence of that regularization rather than a defining property of the momentum mechanism itself. Notably, this interpretation follows directly from the update equations and therefore requires no assumptions regarding stochasticity, convexity, smoothness class, condition number, ergodicity, or dimensionality.

7.2 Momentum is first-order lowpass filter design

The paper shows in Section 2 that the hyperparameters of momentum are the first-order lowpass filter design parameters (β, γ) that satisfy a spectral-norm less than one design condition (6). From this design condition, the popular Heavy-Ball and Nesterov momentum update forms can be realized exactly. As discussed and shown in Section 6, these methods correspond to particular operating points within a broader pole-zero design space (Appendix A), where alternative placements of the filter zero relative to a stable pole are also possible. Also, as show in more detail in Appendix B, these filter-design choices reshape both the stability region and the convergence-rate behavior of the associated mean-error dynamics within the illustrative stochastic error model.

7.3 Motivating modular design principles

Beyond the illustrative stochastic error setup, the rule $\alpha \propto \lambda^{-1}$ can be interpreted as expressing a *normalization principle* that scales parameter update steps inversely with λ, a characteristic measure (or scale) of the gradient magnitude. Under this interpretation, μ is analogous to a trust-region radius [40], and controls the magnitude of the update step.

In general, this principle can be realized in practice via iterative data-driven substitutes $\lambda(t)$ for the fixed scale λ. A general form of the update rule that captures this principle can be expressed as

$$\begin{aligned} \mathrm{v}(t,i) &= \mathbb{H}_{\beta,\gamma}\{\mathrm{g}(t,i)\} \\ \alpha(t) &= \mu\,\lambda^{-1}(t) \\ \mathrm{w}(t+1) &= \mathrm{w}(t) - \alpha(t)\,\mathrm{v}(t). \end{aligned} \tag{46}$$

Here, we assume that $\lambda^{-1}(t)$ is updated during each iteration in some appropriate manner. Interpreting λ in this way cleanly separates momentum (gradient smoothing through a lowpass filter) and learning-rate selection (via a gradient normalization strategy) into modular components. We briefly make this concrete, by showing that the modular design principle in (46) is exhibited in two popular stochastic gradient optimizers: Adam and Muon.

7.3.1 Per-coordinate Second Moment Estimation

For each coordinate $i = 1, \ldots, n$ of the parameter update step $\mathrm{d}(t+1) = \alpha(t)\mathrm{v}(t) = \mu\,\lambda^{-1}(t)\,\mathrm{v}(t)$, choose the normalization scale in the learning-rate vector as

$$\lambda(t,i) = C^{\frac{1}{2}}(t,i), \tag{47}$$

where $C(t,i) = \mathbb{E}[\mathrm{g}^2(t,i)]$. This involves estimating the second moment or power of $\mathrm{g}(t)$ for each coordinate. Consequently, since $\mathbb{E}[\mathrm{d}^2(t+1,i)] = \alpha^2(t,i)\,\mathbb{E}[\mathrm{v}^2(t,i)]$, and from (2.3), $\mathbb{E}[\mathrm{v}^2(t,i)] \le \mathbb{E}[\mathrm{g}^2(t,i)]$. Then, using $\alpha^2(t,i) = \mu^2\,C^{-1}(t,i)$ as defined in (47) yields $\mathbb{E}[\mathrm{d}^2(t+1,i)] \le \mu^2$. The scaling choice (47) can be interpreted as ensuring that under the parameter update step $\mathrm{d}(t+1) = \mu\,\lambda^{-1}(t)\,\mathrm{v}(t)$, each coordinate satisfies the constraint $\mathbb{E}[\mathrm{d}^2(t+1,i)] \le \mu^2$, and therefore it follows that

$$\mathbb{E}\left[\|\mathrm{d}(t+1)\|_2^2\right] \le n\,\mu^2. \tag{48}$$

A simple but practical way to realize (47) is via an exponential moving average (EMA)

$$\mathrm{b}(t,i) = \rho\,\mathrm{b}(t-1,i) + (1-\rho)\,\mathrm{g}^2(t,i), \tag{49}$$

where ρ is a positive constant, close to but less than 1. An additional refinement is to divide the output (49) by $1-\rho^t$ to correct for its transient bias. Finally,

$$\lambda^{-1}(t,i) = 1/\big(\sqrt{\mathrm{b}(t,i)} + \epsilon\big), \tag{50}$$

where ϵ is a small positive constant used to artificially ensure numerical stable computation of the reciprocal of $\sqrt{\mathrm{b}(t,i)}$.

The scaling choice (47) in the learning-rate and its realization above corresponds to Adam (or RMSProp if $\gamma = \beta = 0$) and (48) shows that it directly enforces a trust-region constraint on the expected Euclidean-norm (48) or energy of the update step vector.

7.3.2 Matrix-valued formulation: Matrix Inverse Square-root Estimation

Here, we now extend the parameter update form to a two-dimensional matrix operator-level update form. This operator-level formulation replaces the vectorized learning rate with a linear operator acting on the gradient matrix and replaces scalar second moments with covariance operators. Throughout, we use the Frobenius inner product $\langle A, B\rangle_F = \mathrm{tr}(A^\top B)$, and $\|A\|_F^2 = \langle A, A\rangle_F$, for any matrix $A, B \in \mathbb{R}^{n\times m}$, where $1 < n \le m < \infty$. Let $\mathrm{w}(t),\ \mathrm{g}(t),\ \mathrm{v}(t),\ \mathrm{d}(t+1) \in \mathbb{R}^{n\times m}$, and let the learning rate operator $\alpha(t) \in \mathbb{R}^{n\times n}$ be symmetric and positive definite, such that $\alpha(t) = \alpha(t)^\top$. For this formulation, we will choose the normalization scaling rule as

$$C(t) = \mathrm{v}(t)\,\mathrm{v}(t)^\top, \quad \lambda(t) = C^{\frac{1}{2}}(t), \tag{51}$$

preserved as the matrix square root of a symmetric autocorrelation matrix of the filtered gradient. The inverse scale $\lambda^{-1}(t) = C^{-\frac{1}{2}}(t)$, is the square root of the inverse of a symmetric positive-definite matrix $C(t)$. Therefore, this directly involves estimation of the inverse of the square-root of a matrix. Consequently, $\|\mathrm{d}(t+1)\|_F^2 = \mu^2 \, \|C^{-\frac{1}{2}}(t)\,\mathrm{v}(t)\|_F^2$, and since $\|C^{-\frac{1}{2}}(t)\,\mathrm{v}(t)\|_F^2 = \mathrm{tr}(I_n)$, where $I_n \in \mathbb{R}^{n\times n}$ is an identity matrix, it follows that

$$\|\mathrm{d}(t+1)\|_F^2 = n\,\mu^2, \tag{52}$$

which implies that the scaling choice (51) can be interpreted as satisfying a constant trust-region constraint analogous to the coordinate-wise scaling rule (47). By contrast, (52) shows that the matrix-valued scaling rule (51) directly imposes a trust-region constraint measured in the Frobenius norm.

Substituting (51) into the update step and using the reduced singular value decomposition $\mathrm{v}(t) = U(t)\Sigma(t)V^{\top}(t)$ yields $\mathrm{d}(t+1) = -\mu\,\lambda^{-1}(t)\,\mathrm{v}(t) = -\mu\,\bar{\mathrm{v}}(t)$, where $\bar{\mathrm{v}}(t) = C^{-\frac{1}{2}}(t)\,\mathrm{v}(t) = U(t)V^{\top}(t)$ is the normalized filtered gradient matrix. Since $\|\bar{\mathrm{v}}(t)\|_2^2 = 1$, it follows that

$$\|\mathrm{d}(t+1)\|_2^2 = \mu^2. \tag{53}$$

Thus, in addition to the Frobenius-norm trust-region constraint (52), the normalized update also satisfies a constant spectral-norm constraint (53). A practical way to realize (51) is via the efficient polynomial method introduced by Lakić [41]. This method provides a family of stable, rapidly convergent recursions parameterized by a polynomial of order j, and includes several classical schemes as special cases. During implementation, invertibility of $C(t)$ can be artificially ensured by adding a scaled identity matrix $\epsilon\, I_n$ to $C(t)$. Notably, the classical Newton-Schulz iteration in [42] can be recovered as the special case $j = 2$ of Lakić's polynomial family. Further, a single-step realization of $\mathrm{d}(t+1)$ via the same Newton-Schulz iteration, or a related higher-order polynomial recurrence, directly recovers Muon (Momentum Orthogonalized by Newton-Schulz) [43, 44].

Takeaway: Across scalar, vectorized, matrix-valued forms, the modular principle remains the same. Stochastic gradient optimizers can be designed explicitly by (i) designing a gradient filter and (ii) choosing a learning-rate function corresponding to a trust-region constrained gradient normalization rule.

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Appendix A: Illustration of Pole-zero Placement

More intuitively, one may view the pole-zero first-order filter structure and its frequency response as a terrain surrounding the unit circle in the complex z-plane, annotated by a single stable pole and a single zero inside the unit circle. The frequencies $\omega \in [-\pi, \pi]$ correspond to locations along the circular path $z = e^{j\omega}$, and the magnitude response (gain) $|\mathbb{H}(e^{j\omega})|$ represents the altitude or elevation at each location on this path. Poles act like hills that raise elevations, whereas zeros act like valleys or sinkholes that lower elevations in the landscape. Moving around the unit circle therefore reveals how the gain varies across frequencies, indicating the degree of amplification, preservation, or attenuation associated with each frequency.

Within the design condition (6), specific pole-zero locations reshape this terrain and thereby control the spectral characteristics of the filter. For example, placing a pole near $z = 1$ creates a hill along the portion of the terrain associated with lower frequencies, while a nearby zero creates a valley that lowers the surrounding terrain. The transfer function provides the design map, while the corresponding difference equation implements that design.

Figure 2 illustrates the placement of the pole and zero for the stochastic gradient algorithm without momentum, Polyak's Heavy-Ball and Nesterov Accelerated Gradient.

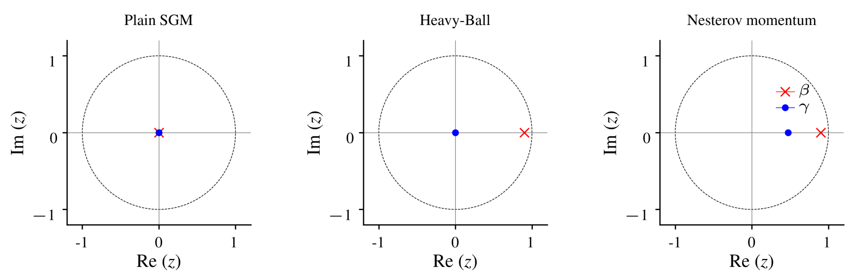

Figure 2: Illustrative Pole-zero locations in the complex z-plane for plain stochastic gradient descent (SGM), Heavy-Ball momentum (HB), and Nesterov momentum (NAG). HB introduces a single stable pole with a zero at the origin, while NAG places a zero inside the unit circle relative to the same pole.

Appendix B: Further Insights from the Illustrative Stochastic Error Setup

Distinct pole-zero configurations induce different transient and stability properties.

Mean-error dynamics for Heavy-Ball momentum

Now consider the filter design corresponding to $\gamma = 0$ and $0 \leq \beta < 1$. This leads to $a_p - k_p = \beta$, and the characteristic polynomial $z^2 - (1 + a_p)\, z + (a_p - k_p) = 0$ simplifies to $z^2 - (1 + a_p)\, z + \beta = 0$ with discriminant $D = (1 + a_p)^2 - 4\,\beta$. Setting $1 + a_p = 1 + \beta - \alpha\eta\lambda = 0$ gives $\alpha\eta\lambda = 1 + \beta$, the characteristic polynomial becomes $z^2 + \beta = 0$ and the discriminant is minimized as $D = -4\beta$. With this choice, the two system eigenvalues of (39) satisfy a common magnitude

$$|z_1| = |z_2| = \sqrt{|\beta|} < 1, \tag{54}$$

which ensures exponential stability with decay rate $\sqrt{|\beta|} < 1$. For the lowpass filter case $\beta > 0$, the eigenvalues form a complex-conjugate pair $z_{1,2} = \pm\, j\sqrt{\beta}$, corresponding to $D = -4\beta < 0$. For the no filtering case $\beta = 0$, the eigenvalues degenerate to repeated values $|z_1| = |z_2| = 0$ yielding the fastest possible expected error decay and $D = -4\beta = 0$. Generally, the characteristic polynomial $z^2 - (1 + a_p)\, z + \beta = 0$ satisfies $z_1 + z_2 = 1 + a_p$ and $z_1\, z_2 = \beta$. Solving the discriminant $D \leq 0$ for α leads to the learning-rate interval

$$\frac{(1-\sqrt{|\beta|})^2}{\eta}\,\frac{1}{\lambda} \leq \alpha \leq \frac{(1+\sqrt{|\beta|})^2}{\eta}\,\frac{1}{\lambda}. \tag{55}$$

The midpoint of this interval $\alpha = \frac{1+\beta}{\eta}\frac{1}{\lambda} = \frac{1+\beta}{1-\beta}\frac{1}{\lambda}$ corresponds to $1 + a_p = 0$. Consequently, as $\beta \to 0$, and $\alpha \to \frac{1}{\lambda}$, then both the plain SGM method and the fastest possible convergence mode is recovered in the limit.

In summary, the filter design choice $\gamma = 0$ and $0 \le \beta < 1$ induces a mean error dynamics whose exponential stability or decay rate depends solely on the pole location β.

Stability Region Shaping as a Function of Filter Design

The characteristic polynomial $z^2 - (1 + a_p)\, z + (a_p - k_p) = 0$ of the expected error system matrix $\mathbf{A}$ in (39) is of the second-order. For a second-order system, convergence rate is defined by the dominant eigenvalue magnitude $\max\{|z_1|, |z_2|\}$. In general, the stable, equal eigenvalue magnitude $|z_1| = |z_2| < 1$ is a necessary minimax optimal condition [2].

The full stability region $|z_{1,2}| < 1$ for the learning rate α or its normalized form $\alpha\eta\lambda$ as a function of the filter pole β can be obtained by applying the Jury stability conditions [45] to this characteristic polynomial, which simplifies to three necessary and sufficient conditions: $1-(1+a_p)+(a_p-k_p) > 0$, $1 + (1 + a_p) + (a_p - k_p) > 0$, and $1 - (a_p - k_p) > 0$.

For Nesterov momentum, the full stability region, using $|\beta| < 1$, is defined by the continuous range,

$$\begin{aligned} -\tfrac{1}{3} \le \beta < 1: &\quad 0 < \alpha\eta\lambda < \tfrac{2(1+\beta)^2}{1+2\beta}, \\ -1 < \beta \le -\tfrac{1}{3}: &\quad 0 < \alpha\eta\lambda < -\tfrac{1-\beta^2}{\beta}. \end{aligned} \tag{56}$$

Next, apply the equal magnitude condition $|z_1 + z_2| \le 2\sqrt{|\beta - \alpha\eta\lambda\gamma|}$ to get a symmetric normalized learning rate $\alpha\eta\lambda$ interval for which the eigenvalues have equal magnitude as a function of β,

$$(1+\beta) - \tfrac{4|\beta|}{1+\beta} \le \alpha\eta\lambda \le (1+\beta) + \tfrac{4|\beta|}{1+\beta} \tag{57}$$

The stable, equal-magnitude region $|z_1| = |z_2| = \sqrt{|\beta - \alpha\eta\lambda\gamma|} < 1$ must lie within the stable region in $|z_{1,2}| < 1$. The learning-rate interval in the stable, equal-magnitude region is defined by

$$\begin{aligned} -\tfrac{1}{3} \le \beta < 1: &\quad \max\Big(0, (1+\beta) - \tfrac{4|\beta|}{1+\beta}\Big) \le \alpha\eta\lambda \le \min\Big(\tfrac{2(1+\beta)^2}{1+2\beta}, (1+\beta) + \tfrac{4|\beta|}{1+\beta}\Big) \\ -1 < \beta \le -\tfrac{1}{3}: &\quad \max\Big(0, (1+\beta) - \tfrac{4|\beta|}{1+\beta}\Big) \le \alpha\eta\lambda \le \min\Big(-\tfrac{1-\beta^2}{\beta}, (1+\beta) + \tfrac{4|\beta|}{1+\beta}\Big). \end{aligned} \tag{58}$$

Similarly, for Heavy-Ball momentum, the full stability region $|z_{1,2}| < 1$ is defined by

$$0 \le \beta \le 1: \quad 0 < \alpha\eta\lambda < 2(1+\beta) \tag{59}$$

Apply the equal magnitude condition $|z_1 + z_2| \le 2\sqrt{|\beta|}$ to get

$$(1 - \sqrt{|\beta|})^2 \le \alpha\eta\lambda \le (1 + \sqrt{|\beta|})^2 \tag{60}$$

The stable, equal-magnitude region in $|z_1| = |z_2| = \sqrt{|\beta|} < 1$ is defined by the interval

$$0 \le \beta \le 1: \quad \max\Big(0, (1 - \sqrt{|\beta|})^2\Big) \le \alpha\eta\lambda \le \min\Big(2(1+\beta), (1 + \sqrt{|\beta|})^2\Big) \tag{61}$$

These stability regions and learning-rate intervals are illustrated in Figure 3, and Figure 4.

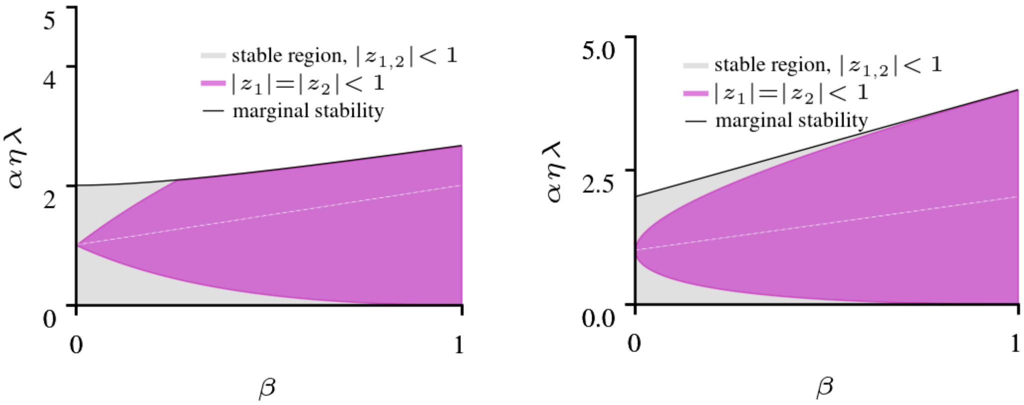

Figure 3: Stability regions in the $(\beta,\ \alpha\eta\lambda)$ plane for **Nesterov momentum (left)** and **Heavy-Ball momentum (right)** given $0 \leq \beta < 1$. Gray shading marks the Jury-stability region where both eigenvalues satisfy $|z_{1,2}| < 1$, the magenta area marks the stricter equal-magnitude region $|z_1| = |z_2| < 1$, and the black curve shows the marginal stability boundary $|z_1| = |z_2| = 1$. For **Nesterov momentum**, the dotted thin white lines inside the magenta-colored area marks the closed-form parameterization for the fastest possible exponential decay rate $|z_1| = |z_2| = \sqrt{|\beta - \alpha\eta\lambda\gamma|} = 0$ of the mean error dynamics. For the **Heavy-Ball momentum** parameterization, the magenta area marks a stricter equal-magnitude region $|z_1| = |z_2| = \sqrt{|\beta|} < 1$ corresponding to its fastest possible exponential decay of the mean-error dynamics, the thin white lines mark the points where $\alpha\eta\lambda = 1 + \beta$. The $(\beta,\ \alpha\eta\lambda)$ plane shows how both design choices expands and reshapes the range of stable learning-rates compared to the single dot ($\beta = 0$) of the plain SGM, reflecting the lowpass regularization effect of momentum.

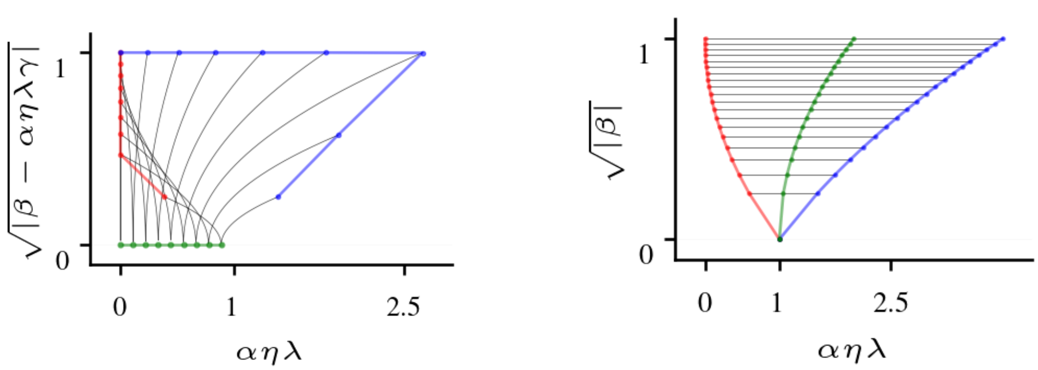

Figure 4: **Left panel: Nesterov momentum** ($\gamma = \beta/(1+\beta)$): Equal eigenvalue magnitude $|z_1| = |z_2| = \sqrt{|\beta - \alpha\eta\lambda\gamma|} < 1$ of the stable expected error system versus the normalized learning rate $\alpha\eta\lambda$. Red and blue dots represent the minimum and maximum points of the stable learning rate intervals for any $0 \leq \beta < 1$. The green dots represent points where the equal-magnitude hits the origin $|z_1| = |z_2| = 0$ which indicate the fastest possible exponential decay rate to 0, and $\alpha\eta\lambda = 1 + \beta$. Plain SGM corresponds to the single dot ($\beta = 0$, $\alpha\eta\lambda = 1$). **Right panel: Heavy-Ball momentum** ($\gamma = 0$): Equal eigenvalue magnitude $|z_1| = |z_2| = \sqrt{|\beta|} < 1$ versus the normalized learning rate $\alpha\eta\lambda$. This represents the fastest possible decay rate for the stable expected error system when $\gamma = 0$. Red and blue dots represent the minimum and maximum points of the stable learning rate intervals for any $0 \leq \beta < 1$. The green dots represent the points where $\alpha\eta\lambda = 1 + \beta$. Plain SGM corresponds to the single dot ($\beta = 0$, $\alpha\eta\lambda = 1$).